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June 6, 2025

*MDM2* Amplification in ecDNA – BIMM/BENG/CSE 182 Final Project Report

## **Part 1**

### ***Introduction - Role of MDM2 in cancer***

In this study, our team investigated the role of the *MDM2* gene in cancer, particularly in the context of extrachromosomal DNA (ecDNA) amplification. *MDM2* (Murine Double Minute 2) is a gene that encodes an E3 ubiquitin ligase, which is responsible for regulating the tumor suppressor protein p53. Under normal physiological conditions, *MDM2* maintains p53 homeostatically at low levels to prevent unwanted cell cycle arrest or apoptosis. However, when *MDM2* is overexpressed/amplified, it can excessively degrade p53, undermining its tumor suppressor functions and facilitating uncontrolled cell proliferation.

*MDM2* amplification is found in multiple human cancers: liposarcomas, breast cancer, osteosarcoma, and glioblastoma. Additionally, *MDM2* expression is upregulated by Estrogen Receptor Alpha (ERα) through a p53-independent pathway. *MDM2* can also degrade ERα via ubiquitination, creating a complex feedback loop that may promote hormone-derived cancer progression, such as breast cancer.

### ***Evolutionary History of MDM2***

Comparative genomics studies have shown that *MDM2* and its paralog *MDM4* originated from a gene duplication event that occurred before the emergence of bony vertebrates, over 440 million years ago.

| | | | | |
|---|---|---|---|---|
| . . Leucoraja erinaceus | sharks & rays | 636 | 4 | Leucoraja erinaceus hits |
| . . Stegostoma tigrinum | sharks & rays | 636 | 9 | Stegostoma tigrinum hits |
| . . Amblyraja radiata | sharks & rays | 623 | 5 | Amblyraja radiata hits |
| . . Callorhinchus milii | chimaeras | 607 | 5 | Callorhinchus milii hits |
| . Myxine glutinosa | hagfishes | 421 | 2 | Myxine glutinosa hits |

**Figure 1:** Protein-protein BLAST results identifying distant orthologs of *MDM2*.

This table shows BLASTP hits for MDM2 against early-diverging vertebrates, including sharks, rays, chimaeras, and hagfish. Notably, the most distantly related hit was found in *Myxine glutinosa* (Atlantic hagfish), with a bit score of 421 and 2 hits.

To support the studies, we performed a protein-protein BLAST search, where we identified one of the most distantly related orthologs of *MDM2* in the Atlantic hagfish (*Myxine glutinosa*), a jawless vertebrate, suggesting functional conservation for at least ~310 million years. Furthermore, homologs of the *MDM* gene family have been detected in invertebrates such as *Ciona intestinalis* (sea squirt), indicating that the ancestral origin of the gene family more than 500 million years.

### *Amplification Types in Pediatric vs. Adult Tumors*

To understand the distribution of *MDM2* amplification mechanisms, we compared amplification types between pediatric and adult (PCAWG) tumor cohorts using data from the Amplicon Repository. Both cohorts exhibit similar patterns across four amplification types: ecDNA, linear, BFB (breakage-fusion-bridge), and complex-non-cyclic. However, ecDNA amplification was more frequent in pediatric tumors (42.3%) compared to adult tumors (26.4%), a difference of approximately 16%. This suggests that the mechanisms of *MDM2* amplification, and possibly its oncogenic effects, are related to age.

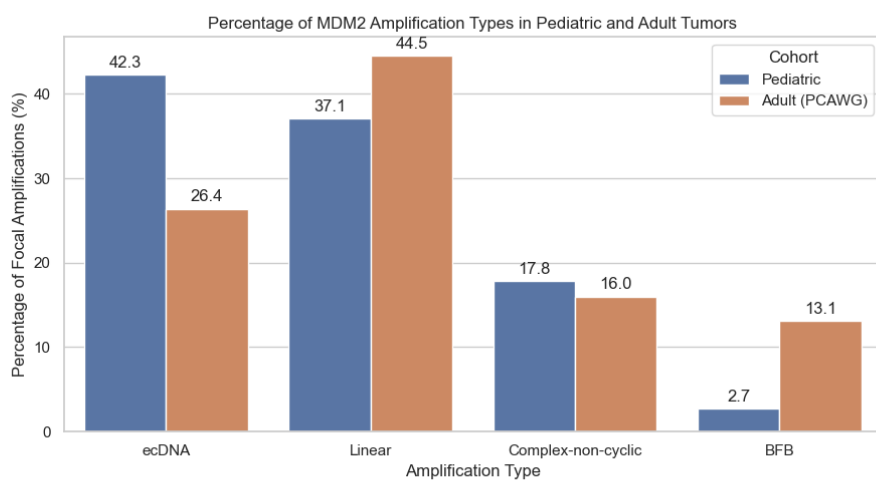


**Figure 2:** Distribution of *MDM2* amplification types in pediatric and adult tumors.
Bar graph comparing the percentage of focal amplifications of the MDM2 gene by amplification type, ecDNA, linear, complex-non-cyclic, and BFB (breakage-fusion-bridge), across pediatric and adult (PCAWG) tumor cohorts.

### *Co-Occurrence of MDM2 with Other Oncogenes on ecDNA*

Our analysis of *MDM2*-containing ecDNA amplicons revealed several oncogenes that frequently co-occur with *MDM2*. In adult tumors, 29 ecDNA amplicons containing *MDM2* were identified, with 68 co-amplified genes. The most common and closely located were *CDK4, YEATS4,* and *AGAP2*. These genes are all located on chromosome 12 and have been implicated in cell cycle regulation and oncogenic signaling. Their recurrent co-amplification with *MDM2* suggests a possible synergistic effect in tumor progression.

In pediatric tumors, 15 *MDM2*-containing ecDNAs were found, with 23 co-occurring genes. The same three oncogenes, CDK4, YEATS4, and AGAP2, were also frequently amplified. The consistent co-localization of these genes across distinct age cohorts implies that they may be part of a core oncogenic module on ecDNA. The preservation of these combinations across tumor types suggests selective advantages conferred by their joint amplification, such as enhanced proliferation, survival, or evasion of cell-cycle checkpoints.

**Table 1:** Genes co-amplified with *MDM2* on ecDNA in adult and pediatric tumors.

<table>
<tr><th>Gene</th><th>Count</th><th>Fraction</th><th colspan="2">Distance to MDM2 (bp)</th></tr>
<tr><td>CDK4</td><td>24</td><td>0.83</td><td colspan="2">11076842</td></tr>
<tr><td>YEATS4</td><td>23</td><td>0.79</td><td colspan="2">548446</td></tr>
<tr><td>AGAP2</td><td>23</td><td>0.79</td><td colspan="2">11094891</td></tr>
<tr><td>HMGA2</td><td>15</td><td>0.52</td><td colspan="2">2931473</td></tr>
<tr><td>IFNG</td><td>13</td><td>0.45</td><td colspan="2">669606</td></tr>
<tr><td>TBC1D15</td><td>8</td><td>0.28</td><td colspan="2">3055176</td></tr>
<tr><td>DDIT3</td><td>7</td><td>0.24</td><td colspan="2">11308307</td></tr>
<tr><td>GLI1</td><td>5</td><td>0.17</td><td colspan="2">11360834</td></tr>
<tr><td>SYT1</td><td>5</td><td>0.17</td><td colspan="2">10331531</td></tr>
<tr><td>ELK3</td><td>5</td><td>0.17</td><td colspan="2">27405239</td></tr>
<tr><td>PA2G4</td><td>4</td><td>0.14</td><td colspan="2">12717626</td></tr>
<tr><td>ERBB3</td><td>4</td><td>0.14</td><td colspan="2">12735025</td></tr>
<tr><td>LRIG3</td><td>4</td><td>0.14</td><td colspan="2">9930531</td></tr>
<tr><td>NACA</td><td>4</td><td>0.14</td><td colspan="2">12107997</td></tr>
<tr><td>WIF1</td><td>3</td><td>0.1</td><td colspan="2">3740895</td></tr>
<tr><td>DCD</td><td>3</td><td>0.1</td><td colspan="2">14180383</td></tr>
<tr><td rowspan="2">Rare MDM2 co-occurrences (≤2 samples)</td><td rowspan="2">52 genes</td><td rowspan="2">0.03~0.07</td><td>∞</td><td>37</td></tr>
<tr><td>Ch12</td><td>14</td></tr>
</table>

<table>
<tr><th>Gene</th><th>Count</th><th>Fraction</th><th colspan="2">Distance to MDM2 (bp)</th></tr>
<tr><td>YEATS4</td><td>14</td><td>0.93</td><td colspan="2">548446</td></tr>
<tr><td>AGAP2</td><td>9</td><td>0.6</td><td colspan="2">11094891</td></tr>
<tr><td>CDK4</td><td>9</td><td>0.6</td><td colspan="2">11076842</td></tr>
<tr><td>IFNG</td><td>7</td><td>0.47</td><td colspan="2">669606</td></tr>
<tr><td>DDIT3</td><td>7</td><td>0.47</td><td colspan="2">11308307</td></tr>
<tr><td>LRIG3</td><td>6</td><td>0.4</td><td colspan="2">9930531</td></tr>
<tr><td>GLI1</td><td>6</td><td>0.4</td><td colspan="2">11360834</td></tr>
<tr><td>HMGA2</td><td>4</td><td>0.27</td><td colspan="2">2931473</td></tr>
<tr><td>TBC1D15</td><td>3</td><td>0.2</td><td colspan="2">3055176</td></tr>
<tr><td>WIF1</td><td>3</td><td>0.2</td><td colspan="2">3740895</td></tr>
<tr><td>CCND2</td><td>3</td><td>0.2</td><td colspan="2">64602824</td></tr>
<tr><td>FGF6</td><td>3</td><td>0.2</td><td colspan="2">64362362</td></tr>
<tr><td>BTG1</td><td>2</td><td>0.13</td><td colspan="2">23316202</td></tr>
<tr><td>LDHB</td><td>2</td><td>0.13</td><td colspan="2">47180268</td></tr>
<tr><td>KRAS</td><td>2</td><td>0.13</td><td colspan="2">43598773</td></tr>
<tr><td rowspan="2">Rare MDM2 co-occurrences (=1 samples)</td><td rowspan="2">8 genes</td><td rowspan="2">0.07</td><td>∞</td><td>6</td></tr>
<tr><td>Ch12</td><td>2</td></tr>
</table>

Tables display the most frequent oncogenes co-occurring with *MDM2* on ecDNA amplicons in adult tumors (left, PCAWG cohort) and pediatric tumors (right). The "∞" symbol indicates that the gene is located on a different chromosome from *MDM2*. Rare co-occurrences are grouped at the bottom of each table.

**Part 2**

***Amino Acid Sequence***

Using the NCBI Refseq database, we obtained the amino acid sequence for *MDM2* transcript variant 1:

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MVRSRQMCNTNMSVPTDGAVTTSQIPASEQETLVRPKPLLLKLLKSVGAQKDTYTMKE
VLFYLGQYIMTKRLYDEKQQHIVYCSNDLLGDLFGVPSFSVKEHRKIYTMIYRNLVVVN
QQESSDSGTSVSENRCHLEGGSDQKDLVQELQEEKPSSSHLVSRPSTSSRRRAISETEENS
DELSGERQRKRHKSDSISLSFDESLALCVIREICCERSSSSESTGTPSNPDLDAGVSEHSGD
WLDQDSVSDQFSVEFEVESLDSEDYSLSEEGQELSDEDDEVYQVTVYQAGESDTDSFEE
DPEISLADYWKCTSCNEMNPPLPSHCNRCWALRENWLPEDKGKDKGEISEKAKLENST
QAEEGFDVPDCKKTIVNDSRESCVEENDDKITQASQSQESEDYSQPSTSSSIIYSSQEDVK
EFEREETQDKEESVESSLPLNAIEPCVICQGRPKNGCIVHGKTGHLMACFTCAKKLKKRN
KPCPVCRQPIQMIVLTYFP

---

This is the most complete isoform of *MDM2* present in the database.

***Genomic Region***

*MDM2* contains 497 amino acids and is located on chromosome 12. The ecDNA sample we chose was sample DO10960-SP23739. The *MDM2*-coding portion of this ecDNA has an origin sequence on chromosome 12 of the GRCh37 genome, ranging from position 69110024 to position 69386333. In order to retrieve the actual sequence, we went to the UCSC Genome Browser and downloaded the nucleotide sequence associated within the given range.

***Genomic Sequence***

We used tBLASTn to BLAST the *MDM2* transcript variant 1 protein sequence against the ecDNA sequence we downloaded from the UCSC Genome Browser. This BLAST uncovered 10 hits. The hits in terms of position ranges on chromosome 12 are as follows:

- 69202987 — 69203076
- 69207371 — 69207412
- 69210590 — 69210748

- 69214104 — 69214154
- 69218143 — 69218211
- 69218336 — 69218431
- 69222551 — 69222712
- 69229610 — 69229768
- 69230453 — 69230530
- 69233043 — 69233627

These hits roughly correspond to 10 exons that make up the coding genomic structure of *MDM2*. Although the BLAST search uncovered 10 exons, *MDM2* transcript variant 1 actually has 11 coding exons, according to the UCSC Genome Browser.

**Human Gene MDM2 (ENST00000258149.11_7) from GENCODE V48lift37**

**Description:** MDM2 proto-oncogene, transcript variant 1 (from RefSeq NM_002392.6)
**Gencode Transcript:** ENST00000258149.11_7
**Gencode Gene:** ENSG00000135679.27_17
**Transcript (Including UTRs)**
**Position:** hg19 chr12:69,201,957-69,239,324 **Size:** 37,368 **Total Exon Count:** 11 **Strand:** +
**Coding Region**
**Position:** hg19 chr12:69,202,258-69,233,629 **Size:** 31,372 **Coding Exon Count:** 11

**Figure 3:** *MDM2* transcript variant 1 as seen on the UCSC Genome Browser.

We initially thought that this discrepancy could be due to *MDM2* being truncated in the ecDNA, but that is unlikely. The coding region in Figure 3 is from position 69202258 to position 69233629, which is well within the 69110024 - 69386333 range on the ecDNA. Additionally, the first exon– the one that BLAST missed– which initially appears quite large, is mostly an untranslated region.

**hg19 ENST00000258149.11_7**

```
>hg19_knownGene_ENST00000258149.11_7_0 range=chr12:69201957-69202257 5'pad=0 3'pad=0 strand=+ repeatMasking=none
GGGCGCGCACCGAGGCACCGCGGCGAGCTTGGCTGCTTCTGGGGCCTGTG
TGGCCCTGTGTGTCGGAAAGATGGAGCAAGAAGCCGAGCCCGAGGGGCGG
CCGCGACCCCTCTGACCGAGATCCTGCTGCTTTCGCAGCCAGGAGCACCG
TCCCTCCCCGGATTAGTGCGTACGAGCGCCCAGTGCCCTGGCCCGGAGAG
TGGAATGATCCCCGAGGCCCAGGGCGTCGTGCTTCCGCGCGCCCCGTGAA
GGAAACTGGGGAGTCTTGAGGGACCCCCGACTCCAAGCGCGAAAACCCCG
G
>hg19_knownGene_ENST00000258149.11_7_1 range=chr12:69202258-69202271 5'pad=0 3'pad=0 strand=+ repeatMasking=none
ATGGTGAGGAGCAG
>hg19_knownGene_ENST00000258149.11_7_2 range=chr12:69202272-69202987 5'pad=0 3'pad=0 strand=+ repeatMasking=none
```

**Figure 4:** *MDM2* transcript variant 1 exons split into individual FASTA sequences.

In figure 4, the first sequence is the UTR, while the second sequence is the actual coding portion of the exon. Since it is so small, it is reasonable that even though it should align with the first 4 amino acids of the *MDM2* protein sequence, the resulting alignment score would be so low that BLAST wouldn't recognize it as a hit.

## Part 3

### *Extracting, Clustering, and Filtering Discordant Edges*

Using the PCAWG database from Amplicon Repository, we extracted 10 datasets containing *MDM2*, listed here: DO10960-SP23739, DO10970-SP23754, DO11447-SP24708, DO13348-SP28335, DO13474-SP28581, DO14049-SP29697, DO218742-SP117983, DO219944-SP121761, DO219945-SP121763, and DO219949-SP121774, which is also accessible on our Google Drive. With these files, we ran a series of Unix commands to extract discordant edges into individual files, then combine them into a single file containing all discordant edges from the 10 relevant amplicons. The Unix commands were as follows, where SAMPLE denotes the sample ID (ex. DO13348-SP28335), xxxxx denotes the long string of characters unique to the extracted results, and X denotes the amplicon number:

1. cd SAMPLE
2. gunzip aa_directory.tar.gz
3. tar -xf aa_directory.tar
4. cd xxxxx…xxxxx_AA_results
5. cat xxxxx…xxxxx_ampliconX_graph.txt | grep "discordant" > SAMPLE_ampliconX_discordant_edges.txt

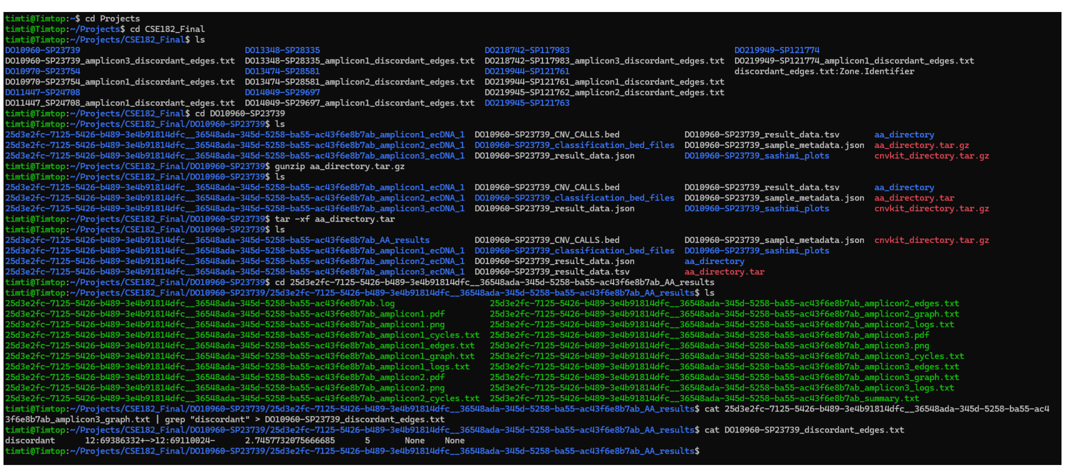

**Figure 5:** Example of using the Commands to Extract Individual Amplicon Discordant Edges

Next, we combined all files into a single "all_edges.txt" file using Unix cat, available here.

To identify common structural variants, we sought to cluster these discordant edges such that if two edges originated and terminated at nearby points on the same chromosome, they would be combined. After trying many cutoffs ranging from 10 to 500,000, we decided on a cutoff of 1,000 base pairs to cluster by, clustering 1298 edges into 1248 clusters. This indicates that most of our edges were unique. Using 10,000 as the cutoff yielded 1227 clusters, which we decided was not enough difference to warrant such a large increase in clustering leniency. Trial clustering cutoff graphs are available here. We wrote the file "cluster_edges.py" to cluster our edges, available here. We decided to filter the resultant clusters such that only edges originating or terminating close to the *MDM2* region of the genome were analyzed in future parts. We decided on an arbitrary broad cutoff of 12:69,000,000 to 12:69,300,000 to analyze.

***Identification of Structural Variant Type***

Next, we proceeded to parse the filtered, clustered edges to identify structural variant types corresponding to each discordant edge. In total, there were 32 inversions, 16 circularizations, 6 translocations, and 17 deletions found from our 71 filtered edges. They were classified in the following way:

- If there was a “+” for the start position of the edge and “-” for the end position, this indicates either a translocation or deletion is present.
    - If the *chromosomes* of the start and end positions are different from one another, the edge would be assigned as a translocation. Otherwise, it would be a deletion.
- If the signs of the start and end positions of the discordant edge were the same, this corresponds to an inversion. The inversion could be either a right foldback if the positions were both “+”, or a left foldback if they were both “-”.
- Lastly, an edge was considered a circularization if there was a “-” at the start position and “+” at the end.

Without filtering, the sashimi plot was very cluttered, and contained edges far from *MDM2*.

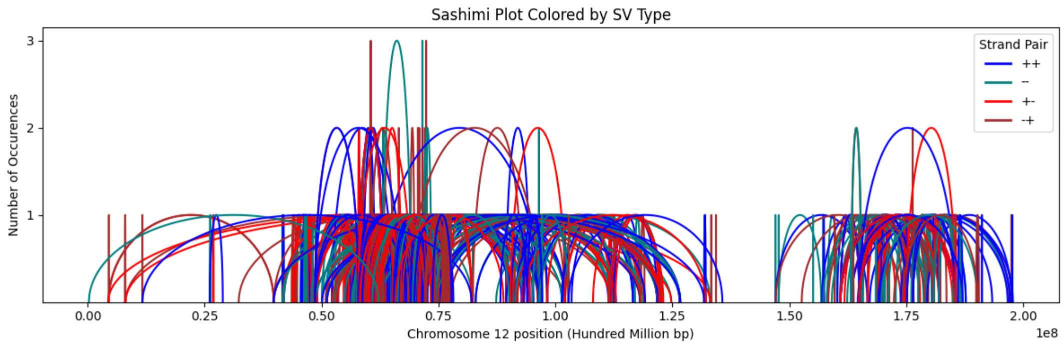


**Figure 6**: Sashimi Plot without Filtering

However, after filtering, a cleaner graph was produced, leaving us with a few relevant edges to analyze. We filtered with the file "filter_edges.py", available here, which filtered out all edges that did not start or end in the desired region of chromosome 12.

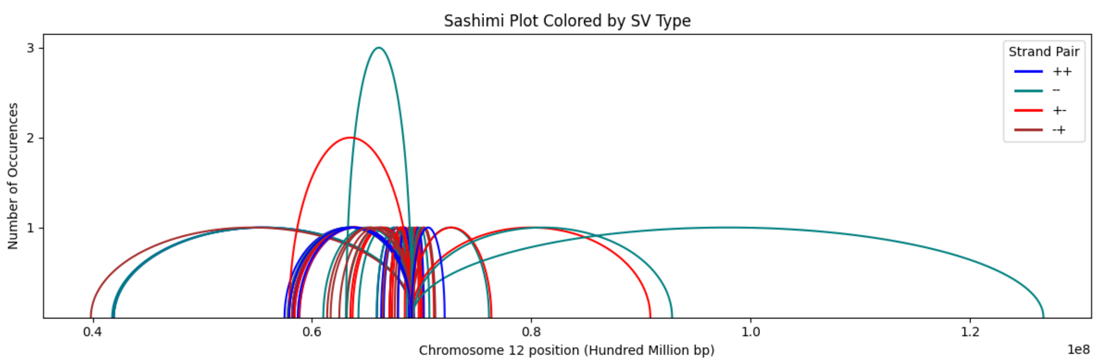


**Figure 7**: Sashimi Plot after Filtering 69,000,000-69,300,000

These plots were made with the file "plot_graph.py", also here. It uses matplotlib to plot arcs, where the height denotes the edge count (# of edges in cluster), the color denotes the SV type, and the endpoints denote the positions on chromosome 12.

### *Biological Relevance of Most Frequently Occurring Structural Variants*

We looked more closely into the biological relevance of the top two most frequently occurring structural variants, with the most common being an inversion (left foldback) that occurred three times around 12:69080680- → 12:63111863-. Through studying this region on UCSC Genome Browser, we found that the start position is within the middle of the *PPM1H* gene, and that the end position is directly within the *NUP107* and *RAP1B* genes. *PPM1H* is known to play a role in tumor suppression through dephosphorylating proteins involved in cell proliferation and metastasis. Meanwhile, *RAP1B* is a small GTPase that can affect cancer growth and invasion through the regulation of cell adhesion, growth, and differentiation. Lastly, *NUP107* encodes a nucleoporin that serves as a component of the nuclear pore complex, which helps to transport all molecules leaving and entering the nucleus. Its overexpression can increase cancer cell survival by making cells more resistant to oxidative damage.

An inversion occurring at these coordinates results in the truncation of these genes in the *MDM2*-containing ecDNA, which could potentially lead to a loss of function or otherwise alter the typical expression levels of these genes. Inhibited *PPM1H* function in the ecDNA would directly promote towards tumor growth due to inactivated suppression; however, in the case of *RAP1B* and *NUP107* which have documented oncogenic functions, it may be possible that their expression is otherwise unaffected if the inversion did not truncate essential parts of the genes' structures required for their expression. Another possibility could be that the inversion brought enhancers or other distal regulatory elements in greater proximity to these genes in the ecDNA to lead to their enhanced expression.

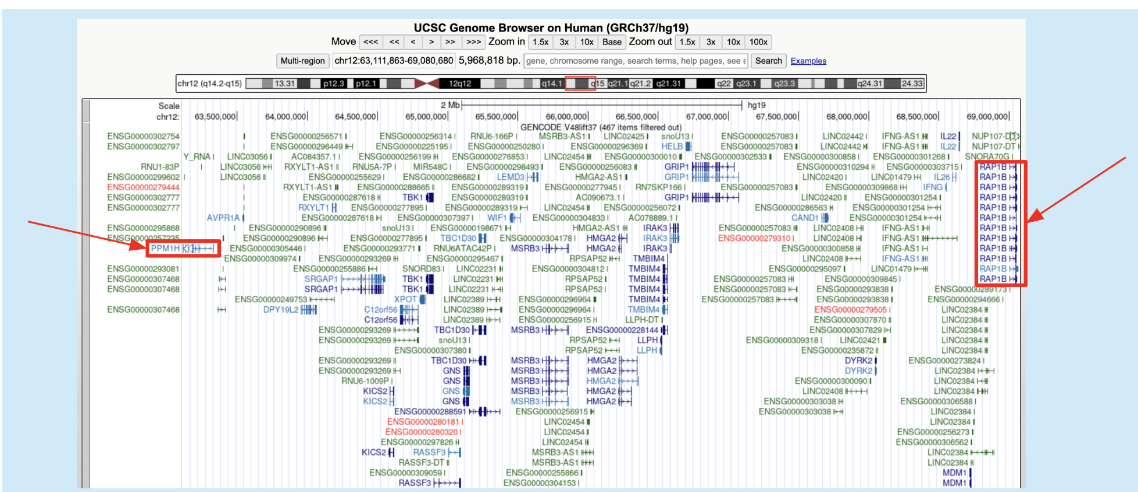


**Figure 8**: Screenshot of the inversion region in UCSC Genome Browser, with *PPM1H* and *RAP1B* outlined in red.

The second-most frequent structural variant was a deletion that occurred two times from 12:69080734+ → 12:57914108-. The deleted region was several million base pairs long and contained 119 genes overall, including *PPM1H* again and additionally the *USP15*, *RASSF3*, and *IRAK3* genes. The latter three genes of interest all have tumor suppressive functions, such as *USP15* through the inhibition of the Wnt signaling pathway, or *RASSF3* through the ubiquination and degradation of *MDM2* to increase *p53* levels. Meanwhile, there is also literature supporting *IRAK3*'s tumor suppressor role in prostate cancer cells. The deletion of all these genes in the ecDNA could heighten tumorigenicity by removing suppressive roadblocks to proliferation in place of more encouraged cell growth. Furthermore, *STAT6* is a transcription factor crucial for the regulation of cell proliferation, metastasis, and apoptosis. It resides directly upstream of the deleted region, while *MDM2* is directly downstream. In the ecDNA, these two genes would likely come into close proximity with one another, and therefore would be much more likely to have interactions between them that promote *MDM2*'s expression.

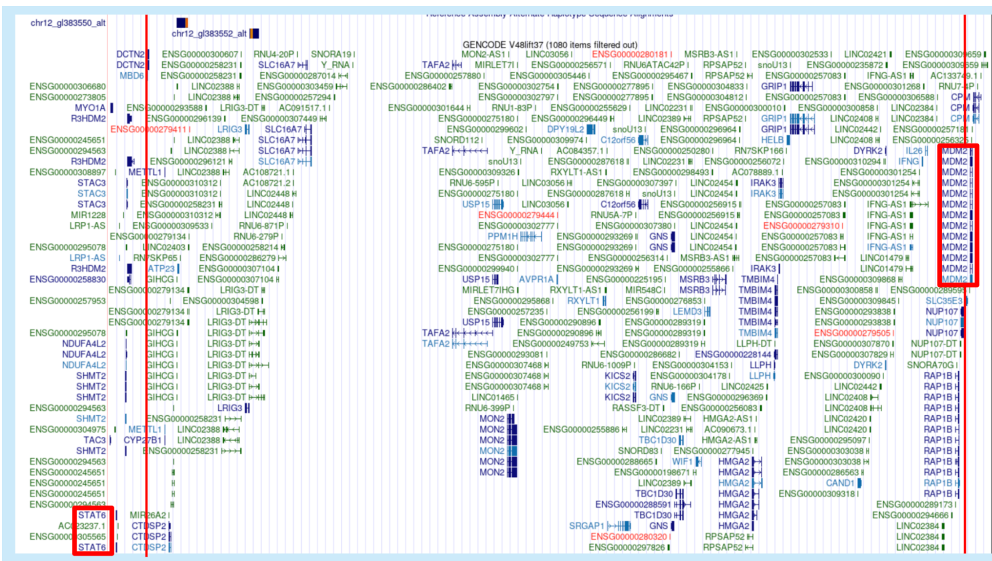

**Figure 9**: Vertical red lines indicating the boundaries of the deleted region in the ecDNA, with *STAT6* and *MDM2* being directly upstream and downstream (respectively) of this region.

Taken altogether, these findings suggest that the change in expression for genes located within the most common structural variants, particularly *PPM1H*, in conjunction with *MDM2* expression in ecDNA may strongly promote cancer progression. In particular, it is the loss of function of tumor suppressor genes and the possible rearrangement of distal regulatory elements to enhance oncogene expression which promotes this outcome. Furthermore, the deletion of genomic regions can bring upstream oncogenes such as the transcription factor *STAT6*, to be in closer proximity to *MDM2* and induce its expression as previously found in the literature.

Citations


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